

Alwyn Mathew

COMPUTER VISION EXPERT — GLOBAL TALENT — ILR

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Last updated on October 15, 2025

WORK EXPERIENCE

University of Cambridge, England, United Kingdom
Research Associate (Mobile mapping)
Construction Information Technology Laboratory

Feb 2022–present

- Designed and deployed a **multi-sensor 3D scanning system** for infrastructure inspection, decreasing cost with cheaper sensors.
- Optimized **SLAM & odometry algorithms** for navigation in large-scale construction sites.
- Built an automated **progress monitoring pipeline** from 3D scans to digital construction models.
- Implemented **object detection & segmentation software** integrated with a graph database API for real-world construction data.

University of Dundee, Scotland, United Kingdom
Postdoctoral Research Assistant (3D Computer Vision)
Division of Imaging Science and Technology

Dec 2021–Jan 2022

- Designed **depth estimation networks** for monocular and wide-FOV endoscopy, enabling real-time 3D reconstruction.
- Developed real-time **lumen detection model** for autonomous colonoscopy.
- Integrated deep learning perception modules with **robotic endoscopy systems**.
- Optimized pipelines for **low-latency inference** on target machine.

Indian Institute of Technology (IIT), Patna, India
Doctoral Research (3D Computer Vision)
Department of Computer Science and Engineering

July 2016–Aug 2021

- Developed novel **monocular and wide-FOV depth estimation models**, enabling real-time 3D scene understanding with a single camera.
- Built a self-supervised **training pipeline** for depth estimation, reducing the need for costly labelled data.
- Investigated **adversarial vulnerabilities** of depth models, contributing to more robust deployment of vision systems in safety-critical settings.

SKILLS

Programming	Python, C/C++
Deep Learning Frameworks	Pytorch, TensorFlow, Keras
Computer Vision	SLAM, 3D Reconstruction, Depth Estimation Object Detection, Segmentation
Tools & Platforms	Git, Docker, ROS
Soft Skills	Project leadership, cross-functional collaboration

EDUCATION	Ph.D. in 3D Computer Vision <i>Indian Institute of Technology (IIT) Patna, Bihar, India</i>	<i>Aug 2021</i>
	Master of Technology in Image Processing (M.Tech) <i>College of Engineering Karunagappally, Kerala, India</i>	<i>May 2016</i>
	Bachelor of Technology in Information Technology (B.Tech) <i>Tamilnadu College of Engineering, Coimbatore, India</i>	<i>April 2012</i>

RECENT PROJECTS

Digital Road Local Authority - Digitizing local roads [Link](#)

- Create detailed mesh to enable the integration of Local Roads into the Digital Roads initiative that focus on Highways.
- Lead data collection with multi sensor scanner and processing to capture odometry of the scan path.
- Developed automated data processing pipeline, reducing manual post-processing time and enabling faster integration into the Digital Roads system.

D-HYDROFLEX - Digitizing Hydropower Plants [Link](#)

- Developing algorithms to automatically generate geometric digital twins of dams, improving asset modeling accuracy and efficiency.
- Automated generation of the relationship graph from point cloud.
- Optimized data processing and mesh generation, enabling rapid simulation and sustainability assessment of hydropower assets.

OMICRON – Digital Building Twin Platform for Road Assets [Link](#)

- Developed a vehicle-mounted mobile scanning system to capture mid-fidelity 3D geometry of road assets.
 - Designed automated data processing pipelines for generating digital twins from captured scans, reducing manual effort.
 - Optimized sensor fusion and SLAM algorithms to ensure accurate 3D reconstruction of road networks.
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